

Monthly Assessment – Python

Topics: Python Basics, Operators, List, Tuple, Dictionary, Set, Conditional Statements & Loops

Section A – Descriptive Questions

Q1. Explain the difference between List, Tuple, Set, and Dictionary in Python.

Mention at least **two differences** for each and give **one suitable example** of each data type.

Q2. Explain the difference between break and continue statements in Python.

Write a small example program for each statement and explain the output.

Q3. Explain the difference between for loop and while loop.

Also mention:

- When a for loop is preferred
- When a while loop is preferred
- What can happen if the condition of a while loop never becomes False

Q4. Consider the following dictionary:

```
student = {  
    "name": "Rahul",  
    "age": 21,  
    "marks": [75, 82, 91],  
    "subjects": ("Python", "Java", "SQL")  
}
```

Answer the following:

- a) How will you access the student's name?
- b) How will you access the second mark from the marks list?
- c) How will you access the first subject from the subjects tuple?
- d) How will you add a new key "city" with value "Pune"?
- e) How will you change the age to 22?

Q5. Explain the difference between the following operators with suitable examples:

1. Arithmetic operators
2. Relational/Comparison operators
3. Logical operators
4. Assignment operators
5. Membership operators

Also explain the output of:

```
x = 10
```

```
y = 5
```

```
print(x > y and x != 5)
```

```
print(x < y or y == 5)
```

```
print(not(x == 10))
```

Q6. Employee Attendance Analyzer – Hard

Write a Python program to store the attendance status of **10 employees** in a list.

The list should contain only:

```
"P" # Present
```

```
"A" # Absent
```

```
"L" # Leave
```

The program should:

1. Count the total number of Present, Absent, and Leave employees.
2. Calculate the attendance percentage using:

Attendance % = Present / Total Working Days × 100

For this question, assume **10 working days** for each employee.

3. Display employees whose attendance is below **75%**.
4. Store employee names and their number of present days in a dictionary.
5. Find the employee with the **highest number of present days without using max()**.
6. Display:

Employee Name

Present Days

Attendance Percentage

Attendance Status

where:

- $\geq 75\%$ → Eligible
- $< 75\%$ → Low Attendance

Concepts expected:

List, Dictionary, for loop, if-elif-else, comparison operators, variables and nested logic.

Q7. Shopping Cart Analyzer – Hard

Write a Python program to create a simple shopping cart.

Store the products and their prices in a dictionary:

```
products = {  
    "laptop": 50000,  
    "mobile": 25000,  
    "headphones": 2000,  
    "keyboard": 1500,  
    "mouse": 800  
}
```

Ask the user to enter products one by one.

The program should:

1. Continue accepting products until the user enters "done".

2. Check whether the entered product exists in the dictionary.
3. If the product does not exist, display "Product not available" and continue accepting the next product.
4. Store selected products in a list.
5. Calculate the total bill.
6. If the total bill is:
 - $\geq 50000 \rightarrow 15\%$ discount
 - $\geq 30000 \rightarrow 10\%$ discount
 - $\geq 15000 \rightarrow 5\%$ discount
 - Below 15000 $\rightarrow$ No discount
7. Display:
 - Selected products
 - Original bill
 - Discount amount
 - Final bill
8. If the user enters "exit" at any point, stop the program immediately using break.

Additional condition:

The program should not allow the same product to be added more than **once**.
If the user enters the same product again, display:

Product already added.

Concepts expected:

Dictionary, List, while, if-elif-else, break, continue, operators and membership operators (in).